IaaS vs PaaS vs SaaS

Quick Definition

Model	What is it?	Who Manages What?
IaaS	Virtual computers & storage in the cloud	You manage: Apps, OS, Database Provider manages: Servers, hardware
PaaS	Ready development environment	You manage: Application code Provider manages: Everything else
SaaS	Ready-made software apps online	You manage: User data Provider manages: Everything

1. IaaS - Infrastructure as a Service

Simple Analogy: Renting an empty office building

What You Get: Virtual computers, storage, networking in the cloud

Example: Amazon EC2

- Startup rents 2 virtual computers from Amazon
- Install their own OS (Linux), database, and app
- They control everything about the servers
- Pay only $65/month instead of $4,500/month

Best For: Developers who want maximum control

2. PaaS - Platform as a Service

Simple Analogy: Renting a fully equipped office (furniture included)

What You Get: Ready development platform with tools, databases, hosting

Example: Heroku

- Developer writes code for social media app
- Deploy with one command: git push heroku main
- Heroku automatically handles: servers, databases, scaling, security, updates
- Pay ~$77/month
- Saves hiring $80k/year DevOps engineer

Best For: Developers who want to focus on code, not infrastructure

3. SaaS - Software as a Service

Simple Analogy: Using an app online (nothing to install or manage)

What You Get: Complete ready-made software accessed through browser

Example: Slack

- Company signs up, invites 50 employees
- Instant messaging, file sharing, searchable history
- No setup needed - just use it
- Slack handles: servers, security, backups, updates
- Pay $4,800/year instead of $55,000/year for email server

Best For: Non-technical teams needing ready-made tools

Responsibility Comparison

IaaS: You Handle Provider Handles

[▓▓▓

Feature	IaaS	PaaS	SaaS
Scaling	Manual	Automatic	Automatic
Best For	Custom apps	Quick development	Existing tools

Real-World Examples

IaaS Examples

- **AWS EC2** - Virtual computers
- **Microsoft Azure VMs** - Cloud servers
- **Google Cloud Compute** - Computing resources

PaaS Examples

- **Heroku** - App deployment
- **Google App Engine** - Web apps
- **Firebase** - Mobile/web backend

SaaS Examples

- **Slack** - Team communication
- **Salesforce** - Customer management
- **Microsoft Office 365** - Email, docs, sheets
- **Zoom** - Video conferencing
- **Figma** - Design tool

Which One to Choose?

Use IaaS if:

- ✅ Building custom application
- ✅ Need maximum control
- ✅ Have DevOps expertise
- ✅ Building machine learning systems

Use PaaS if:

- ✅ Want to focus on coding
- ✅ Building web/mobile apps

- ✅ Need rapid deployment
- ✅ No DevOps expertise

Use SaaS if:

- ✅ Need ready-made software
- ✅ Entire team needs access
- ✅ Want zero setup
- ✅ No technical knowledge needed

Cost Example: Small Startup

IaaS (AWS): $200/month

PaaS (Heroku): $100/month ← Cheapest

SaaS (Slack): $40/month

Combined Best Solution: $340/month (Use all three!)

- IaaS for website infrastructure

- PaaS for APIs

- SaaS for communication & tools

Quick Summary Table

IaaS = "Raw Ingredients" (you cook everything)

PaaS = "Cooking Equipment" (equipment provided, you cook)

SaaS = "Ready Food" (just eat, nothing to prepare)

Most companies use a mix of all three:

- IaaS for infrastructure
- PaaS for specific services
- SaaS for business tools (Slack, Office 365, Salesforce)